Making every real estate decision faster and smarter

DATATHON 2026

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The Challenge

User queries mix hard constraints with soft preferences

Your system needs to separate these two layers and combine them effectively.

QUERY 1

"**3-room** *bright* apartment in **Zurich under 2800 CHF** with **balcony**, *close to public transport*"

■ Hard ■ Soft

QUERY 2

"*Bright family-friendly* flat in **Winterthur**, *not too expensive*, *ideally with parking*"

Mostly soft preferences — harder to rank

QUERY 3

"*Modern* **studio** in **Geneva** for **June move-in**, *quiet area*, *nice views if possible*"

Conflicting: cheap vs. central, quiet vs. views

■ **Hard constraints** = must be respected ■ **Soft preferences** = should influence ranking

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Your Task

Map a natural-language query to a ranked list of listings

STEP 1
Extract Hard Filters
Parse the query into strict criteria: rooms, price, city, features

→

STEP 2
Retrieve Candidates
Query listings that satisfy all hard constraints via the API

→

STEP 3
Rank by Relevance
Score soft preferences and sort by overall relevance

→

OUTPUT
Ranked Results
Sorted listing IDs with scores + explanations

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The Data

Listing Data Structure

IDENTITY & PROPERTY

listing_id 36272162
title "3.5 Zimmer Wohnung"
description full text
category "Wohnung"
marketing_type RENT
status ACTIVE / INACTIVE

LOCATION & GEO

street "38 Röhrliberg"
city "Cham"
postal_code "6330"
canton "ZG"
latitude / longitude 47.186 / 8.451

PRICING & SIZE

price 2450 (CHF/month)
num_rooms 3.5
living_area 97 m²
floor 2
available_date 2025-12-12
construction_year 1979

FEATURES (boolean flags)

HasBalconies ✓ HasBaths ✓ HasParkingIndoor ✓ HasWashingmachine ✓
HasElevator HasGarden HasPool HasFireplace PetsAllowed …

Structured flags where available — not all listings have complete feature data

MEDIA

image_urls array of CDN links
local image bundle

Images useful for multimodal ranking

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Getting Started

Your Starter Kit

https://shorturl.at/NRPAw

HARD FILTER API

Pre-built filtering service. Extract constraints from queries and filter listings.

rooms, price, city, features

AI SDK APP

Scaffolded app with Claude **(provided by Datathon)** for LLM-powered query parsing.

Claude 4 Sonnet ready to go

LISTING DATASET

Swiss rental listings hosted on AWS **(provided by Datathon)**. Structured data + images.

~thousands of listings on S3

IMAGE BUNDLE

Property photos for multimodal analysis. Room detection, quality scoring, visual ranking.

jpg, multiple per listing

DOCKER & COMPOSE

Full local dev environment. One command to spin up all services.

docker compose up

MCP SERVER

Model Context Protocol server for tool-augmented AI agent workflows.

connect your AI to live data

💡 We recommend using **AWS** and **Claude** provided by Datathon — credits are included for all participants.

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DATA ENRICHMENT

Go Beyond the Dataset

Listings have basic data — enrich them with external sources to unlock stronger ranking signals.

Transport & Commute

- SBB API for travel times
- Public transport stops nearby
- Commute duration to key hubs
- Proximity to train stations

Amenities & Services

- Supermarkets, pharmacies, schools
- Restaurants & nightlife density
- Healthcare facilities nearby
- OpenStreetMap POI data (Nominatim)

Geospatial Features

- Geocoding via Nominatim
- Neighborhood classification
- Walkability & bike scores
- Elevation & noise levels

Neighborhood Context

- Average rents in the area
- Safety & crime statistics
- Demographic profiles
- Green space & parks ratio

Image Analysis

- Room brightness & natural light
- Modernity of interiors
- View quality from windows
- Space perception & layout

Text Enrichment

- Extract features from descriptions
- Sentiment of listing language
- Keyword tagging (cozy, modern)
- Multilingual normalization

EXAMPLE — WHY ENRICHMENT MATTERS

"I want a quiet apartment **near ETH** with a **short commute to the city center**, **modern-looking** inside, with **good schools nearby**"

→ **near ETH** needs Geospatial **short commute** needs Transport **modern-looking** needs Image Analysis **good schools** needs Amenities

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Example 1: Clear Hard Constraints

User Query

"**3.5-room** bright apartment in **Zurich** under **CHF 2800** with **balcony**"

Bold = hard filter Orange = soft preference

Candidate Listings After Hard Filters (city = Zürich, rooms = 3.5, price ≤ 2800, balcony = true)

Helle 3.5-Zi an der Wengistrasse

Wengistrasse 2, 8004 Zürich

CHF 2,650/mo

3.5 rooms · 85 m² · 2nd floor

Balcony ✓ Minergie ✓ Washer ✓

Bright Minergie-certified apartment in Kreis 4. Large south-facing windows, modern kitchen, parquet floors.

⭐ **bright + Minergie + central**

Moderne Wohnung Oerlikon

Käferholzstrasse 42, 8057 Zürich

CHF 2,450/mo

3.5 rooms · 75 m² · 3rd floor

Balcony ✓ Washer ✓ Parkett ✓

Maisonette near Bucheggplatz. Rooftop terrace, close to ETH Hönggerberg campus. Well connected.

🏠 **maisonette + terrace**

Gemütliche Wohnung Seebach

Hertensteinstrasse 8, 8052 Zürich

CHF 2,200/mo

3.5 rooms · 78 m² · 4th floor

Balcony ✓ Washer ✓ Parking ✓

Private terrace with access from all rooms. In-unit washer/dryer. Near Seebach station. Quiet area.

🌳 **quiet + parking + value**

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Example 1: Ranking Output

How should these rank for "bright"?

#1

Helle 3.5-Zi, Wengistrasse

CHF 2,650 score: 0.94

Why this rank?

- "Helle" (bright) in title — strong signal
- Minergie = large windows, energy-efficient glass
- South-facing, 2nd floor — maximum light
- Modern 2019 build with open floor plan

#2

Moderne Wohnung, Oerlikon

CHF 2,450 score: 0.82

Why this rank?

- Maisonette = windows on two levels
- Roof terrace — skylights likely
- Near ETH campus — well-maintained area
- No explicit “bright” signal in listing

#3

Gemütliche Wohnung, Seebach

CHF 2,200 score: 0.71

Why this rank?

- “Gemütlich” (cozy) ≠ bright
- 4th floor = good light potential
- Best price, but weaker brightness signal
- Parking + terrace still strong amenities

Key insight: "Bright" is a soft preference — your ranking system must interpret description text + metadata to score it

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Example 2: Vague & Soft-Heavy

User Query

"Bright family-friendly flat in **Winterthur**, not too expensive, ideally with parking"

Challenge: Only "Winterthur" is a hard filter. Everything else is subjective. How does the system interpret "not too expensive"?

System Must Interpret

- **"not too expensive"** → below Winterthur median? bottom 30%? Infer from context: 3.5 rooms → ~CHF 1,600–2,450 range
- **"family-friendly"** → 3+ rooms, near schools/parks, quiet area, ground floor or elevator, pets allowed?
- **"bright"** → large windows, high floor, south-facing, or "hell"/"lichtdurchflutet" in description
- **"ideally with parking"** → bonus, not a hard requirement. HasParkingIndoor flag, or mention in description

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Example 3: Conflicting Preferences

User Query

"Modern apartment in **Basel**, cheap but central, quiet with nice views"

Conflict: "Cheap" and "central" are inversely correlated. "Quiet" and "central" often conflict. How does your system balance tradeoffs?

Strategy A

Prioritize Price

2.5-Zi Kleinbasel, Klybeckstr.

CHF 1,350/mo

48 m² · 2nd floor · 1995 build

✓ Cheapest option
✓ Central Kleinbasel location
✗ Street noise (Klybeck is busy)
✗ Older, not “modern”
✗ No views to speak of

Strategy B

Balance All Signals

3.5-Zi Gundeldingen, Güterstr.

CHF 1,950/mo

72 m² · 5th floor · 2021 build

✓ Modern new build
✓ 5th floor = views + quiet
✓ 10 min to SBB by tram
✗ Not the cheapest
✗ “Central” is debatable

Strategy C

Prioritize Lifestyle

3.5-Zi St. Alban, Münsterplatz

CHF 2,800/mo

85 m² · 3rd floor · Renovated

✓ Premium central location
✓ Rhine views, historic quiet
✓ High-end renovation
✗ Not cheap at all
✗ May exceed user budget

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EXAMPLE — IMAGE ANALYSIS FOR RANKING

What Can You Learn from Photos?

Vision models can extract soft ranking signals that text alone misses

☀️ Brightness & Light

Detect natural light levels, window size, south-facing rooms, time-of-day photography bias

✨ Interior Modernity

Classify kitchen/bathroom age, identify recent renovations, detect style (modern vs. traditional)

📐 Space & Layout

Estimate perceived spaciousness, detect open floor plans, measure visual clutter

🏞️ Views & Surroundings

Identify window views (greenery, cityscape, mountain), balcony presence, outdoor spaces

How to use image signals

Combine vision scores with text-based relevance for multimodal ranking. A listing described as "bright" that also shows dark photos should be penalized.

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Approaches

Solution Directions

LLM QUERY DECOMPOSITION

Use LLMs to parse natural language into structured filters + soft preferences. Handle ambiguity, infer missing constraints.

💡 *Prompt engineering is key*

You can also use the MCP App via Claude and ChatGPT to let that model disambiguate and enable multiturn by returning the follow up question via your API first

HYBRID SEARCH

Combine keyword search (BM25) with semantic embeddings. Structured filters narrow candidates, embeddings rank by meaning.

💡 *Best of both worlds*

EMBEDDING-BASED RANKING

Embed queries and listing descriptions into shared vector space. Rank by cosine similarity. Fine-tune for real estate domain.

💡 *Handles soft preferences well*

MULTIMODAL SIGNALS

Use listing images to assess brightness, modernity, views. Vision models can extract features invisible in structured data.

💡 *Images reveal what text hides*

GEOSPATIAL ENRICHMENT

Use lat/lng to compute distances to transit, schools, parks. Enrich listings with neighborhood characteristics.

💡 *Great for "central" or "quiet"*

LEARN-TO-RANK

Train a model on pairwise preferences. Combine all signals (text, features, geo, images) into a unified relevance score.

💡 *Advanced but powerful*

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BONUS

User Preference History

Personalize ranking using past interaction signals. This is optional but will impress the jury.

Interaction Signals

- **Favorites / saves** — explicit positive signal
- **Click-through rate** — which listings get opened
- **Dwell time** — time spent on a listing page
- **Query refinement patterns** — how users narrow down
- **Dismissals / skips** — implicit negative signal

Build a user profile that shifts ranking weights over time. A user who always saves modern apartments should see them ranked higher.

Example Scenario

A user has previously:

- Saved 5 apartments with Minergie label
- Spent 3x more time on listings with balcony photos
- Skipped all listings above CHF 2,500
- Always searched in Kreis 4–6

→ **Inferred profile:**

Eco-conscious, outdoor living, budget-sensitive, urban core preference. Boost Minergie + balcony + central + affordable in ranking weights.

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Judging

Evaluation & Judging

Peer Review – Hidden and public test set

- **Peer feedback** — Other teams evaluate your output
- **Hidden Test queries**
- **Hard-Filter Precision** — Do results actually satisfy hard constraints?

Jury

- **Technical depth** — Quality of architecture and approach
- **Feature width** — Variety of different approaches and considerations
- **Creativity** — Novel ideas, unexpected data usage
- **Demo quality** — Working prototype, clear presentation
- **Failure analysis** — Show where your system breaks

Deliverables (See GitHub README and challenge.md for more infos)

① Working prototype (publicaly reachable API, Repo) ③ Live demo in your presentation ③ Ranking logic

Submission: Live GitHub repo with running code and reachable API link & final presentation

Please read the README and challenge.md of the provided GitHub Repo

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Job Opportunity

Looking for a world-class AI Engineer to join the mission.

AI-Engineer

- Start: asap
- Workload: 40-100%
- Salary: 30-45 CHF/h
- Location: Zurich Accelerator Office

Applications: info@robinreal.ai

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Let’s hack!

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